

QAOA Maxcut currently failing on the latest version of PennyLane and aut

<https://github.com/PennyLaneAI/qml/issues/36>

ROW 110

QAOA Maxcut currently failing on the latest version of PennyLane and autograd #36

New issue

Closed



josh146 opened on Feb 8, 2020

The QAOA maxcut tutorial is currently failing with the following traceback:

```
File "implementations/tutorial_qaoa_maxcut.py", line 208, in circuit
    U_C(gammas[i])
IndexError: index 1 is out of bounds for axis 0 with size 1
```

The issue seems to be passing the parameter `gamma`. At line 246,

```
neg_obj -= 0.5 * (1 - circuit(gammas, betas, edge=edge, n_layers=n_layers))
```

it can be seen that `gamma = Autograd ArrayBox with value [0.00617866 0.00415137]`. However, within the `QNode` at line 202,

```
@qml.qnode(dev)
def circuit(gammas, betas, edge=None, n_layers=1):
    # apply Hadamards to get the n qubit |<= state
    [print(2, i.val) for i in gammas if isinstance(i, qml.variable.Variable)]
```

we have `gamma = [0.006178660018338691]` - the second element is 'missing'.



josh146 added `bug` on Feb 8, 2020



antalszava on Feb 11, 2020

Contributor

The problem seems to be related to the fact that there is only one `QNode` that is being created throughout the entire execution of the script.

When calling `_construct` with two parameters for `gamma`, the same `self.arg_vars` will be used as created for only one parameter (as it is not `None`), yielding the error.

A quick way to check this would be commenting out lines `bitstrings1 = qaoa_maxcut(number_of_layers=1)[1]` and `plt.hist(bitstrings1, bins=bins)`.

Going forward with trying to figure out why there's only one `QNode` that is being created.



antalszava on Feb 11, 2020

Contributor

With one `QNode` the logic should be altered here such that there is a more in-depth check comparing the current `self.arg_vars` and the new `Variables`:

<https://github.com/XanaduAI/pennylane/blob/719e27c7d046e7d488b8c68685fe803f4f5afa9f/pennylane/qnodes/base.py#L500>



josh146 on Feb 11, 2020

Member Author

Thanks @antalszava! Was this bug introduced with the circuit drawer then? We should add a test to PL to test this specific edge case, and make sure it is fixed.



Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

BaseQNode._construct fix
PennyLaneAI/pennylane

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Participants



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antalszava on Feb 11, 2020

Contributor ...

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Participants



antalszava on Feb 11, 2020

Contributor ...

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Member Author ...

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1



antalszava mentioned this on Feb 12, 2020

🔗 [Fix multiple evaluations of mutable QNode PennyLaneAI/pennylane#505](#)

🔗 [Include unit tests for auxiliary methods of BaseQnode PennyLaneAI/pennylane#506](#)



smite mentioned this on Feb 14, 2020

🔗 [BaseQNode._construct fix PennyLaneAI/pennylane#510](#)



antalszava on Feb 18, 2020

Contributor ...

With [PennyLaneAI/pennylane#505](#) merged, I'll close this for now.



🔒 antalszava closed this as **completed** on Feb 18, 2020

Fix multiple evaluations of mutable QNode #505

<> Code

Merged josh146 merged 18 commits into master from mutable_qnode_make_variable on Feb 18, 2020

Conversation 31 Commits 18 Checks 0 Files changed 3

+57 -3



antalszava commented on Feb 12, 2020 · edited

Contributor

Context:

Currently, a mutable `QNode` evaluated multiple times fails if there is a change in the list `Variable` instances that are being used. This can be seen currently when running the QAOA Maxcut tutorial.

Description of the Change:

Adds a slightly more sophisticated check for when to generate the list of `Variable` instance in the `BaseQNode._construct` method.

Benefits:

A very lightweight check is being used for mutable `QNode`. This assumes that the input `args` argument can be converted to a `numpy.ndarray` and that the `shape` attribute of this `numpy.ndarray` will uniquely identify the arguments passed to `_construct`. The use of methods from the `inspect` library is deliberately avoided such that no considerable overhead is introduced.

The test suite passes including a new Integration test getting the edge case similar to the one in the QAOA MaxCut tutorial.

Possible Drawbacks:

This PR addresses the case of when the `QNode` is mutated in its args in a specific variety of ways (querying the shape of a `numpy.ndarray`).

Further TODO items are contained in the following issues:

[#506](#) (this might also contain an unexpected behaviour when using keyword arguments)
[#507](#)

Related GitHub Issues:

[PennyLaneAI/qml#36](#)



Reviewers

- josh146 ✓
- johannesjmeyer ●
- smite ●

Assignees

No one assigned

Labels

bug review-ready

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

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antalszava added 5 commits 5 years ago



First integration test (catches the index out of range error for muta...

d553ea1



Revert logic and Qnode Base integration test

7c77571



Add args_changed check for mutable QNodes

b277b92



Modify construction condition

8862e88



Docstring

1549cb8

4 participants

